



Cyber Attacks Explained

Network Sniffing

Part—3

In earlier articles in this series, we have explored denial of service attacks and packet spoofing. This month, let us focus on a wider hacking approach called network sniffing, which is very commonly used by prying hackers. We will look at a few tools typically used as sniffers and also figure out how to protect IT infrastructure from such attacks.

Sniffing involves capturing, decoding, inspecting and interpreting the information inside a network packet on a TCP/IP network. The purpose is to steal information, usually user IDs, passwords, network details, credit card numbers, etc. Sniffing is generally referred to as a 'passive' type of attack, wherein the attackers can be silent/invisible on the network. This makes it difficult to detect, and hence it is a dangerous type of attack.

As we already learnt over the previous months, the TCP/IP packet contains vital information required for two network interfaces to communicate with each other. It contains fields such as source and destination IP addresses, ports, sequence numbers and the protocol type. Each of these fields is crucial for various network layers to function, and especially for the Layer 7 application that makes use of the received data. By its very nature, the TCP/IP protocol is only meant for ensuring that a packet is constructed, mounted on an Ethernet packet frame, and reliably delivered from the sender to the receiver across networks. However, it does not by default have mechanisms to ensure data security. Thus, it becomes the responsibility of the upper network layers to ensure that

information in the packet is not tampered with.

To understand why hackers sniff, we need to know what they can get from the network. Figure 1 shows the OSI layers and the information a hacker can steal at each layer by successfully sniffing a network.

The sniffing process is used by hackers either to get information directly or to map the technical details of the network in order to create a further attack. Hackers are always in favour of sniffing, because it can be done for a longer time without getting caught.

How do they 'sniff'?

Network sniffing uses sniffer software, either open source or commercial. Broadly, there are three ways to sniff a network, as shown in Figure 2. It is important to remember that sniffing can range from Layer 1 through Layer 7. Talking about physical connectivity, a person (who may be an employee of the firm) who is already hooked up to the internal LAN can run tools to directly capture network traffic. Using spoofing techniques, a hacker outside the target network can intercept packets at the firewall level and steal the information. In the latest form of packet